

Fulfilling Specific Competency 1: Demonstrate real-world attacks over computer networks and configure suitable defenses on network devices to mitigate the attacks.

Due Dates:

Report + Program: Sunday 23 Jun 23:59, submission via **LMS ICT2502C Dropbox**.

Demo: Week 8 Monday 24 Jun, timeslot to be arranged.

PART 1: Investigating and Demonstrating Real-World Network Attacks and Defenses

1.1 Introduction

You have experimented with various network attacks and defenses in ICT2502C lab exercises. Hope that you have enjoyed, and most importantly have a better understanding of how networks can be attacked and defended in the real world.

Of course, there are more network attacks and defenses that we do not have time to cover them all in the lab exercises. So with the foundations that you've learnt, now is your opportunity to **investigate and design ONE lab exercise** on any network attacks and defenses of your interest, and to **demonstrate your knowledge and skills** in conducting in the lab environment.

1.2 Learning Outcomes

Upon completion of this assignment, you should be able to:

- Investigate and design real-world network attacks and defenses of your interest
- Demonstrate your skills in conducting the attacks that you've investigated (You may use any existing tools in Kali Linux, and bonus if using a security tool to be developed by you in Part II.)
- Implement defenses against the attacks that you have demonstrated (If no suitable defense is available, justify in your report.)

1.3 Requirements

Your attacks and defenses must be done over a network. For example, you may:

- Extend network attacks and defenses that you've learnt in the lab exercises; or
- Investigate real world network attacks and defenses that has caught your interest but was not covered; or
- Design and experiment with your own network attacks and defenses

As the saying goes – the sky's the limit! Or does the sky have a limit? Enjoy!

PART 2: Programming Security Tool using Python with Scapy (Bonus for better grade)

2.1 Introduction

You have been using the security tools in Kali Linux for your lab exercises. However, to be IT security professional, we cannot behave like script kiddies who just wait for someone to program the tools that we need. Hence in this Part II, you are going to try programming a security tool yourself!

Programming security tools requires the ability to manipulate the values in the protocol headers to send out malicious frames/packets to attack. Nowadays, manipulating the values in the protocol headers is made simpler with the availability of add-on packages/libraries, e.g. Scapy in Python.

For an introduction to Scapy, please read:

Download Scapy: <https://scapy.net/>

Scapy's documentation: <https://scapy.readthedocs.io/en/latest/>

Build your own tools with Scapy:

<https://scapy.readthedocs.io/en/latest/extending.html>

2.2 Learning Outcomes

Upon completion of this assignment, you should be able to:

- Design a security tool based on the theoretical understanding of how network attacks work
- Program a security tool using Python with Scapy

2.3 Requirements

Depending on the design of your attacks in Part I, you may be using one or more security tools in Kali. Instead of using all the existing tools in Kali, your program must:

- Be used to conduct **at least ONE attack** in Part I

Note: No mark will be awarded for GUI. So your program may simply be run using command line since this is a module on Network Security, and not HCI.

Submission and Grading for PART 1 and 2:

Report covering:

- Theoretical explanation of your network attacks and defenses
- Instructions on your lab network topology setup and procedure to demonstrate the attacks and defenses
- Relevant Wireshark captures to show your program is indeed sending out the malicious attack frames/packets as intended
- Relevant screen shots as evidences to show your network attacks and defenses are successful

Demo covering:

- Practical setup of your lab network environment and successful demonstration of your network attacks and defenses

Text files containing:

- Source codes of your program

Grading scheme:

- How realistic and creative are your network attacks and defenses
- Theoretical understanding of how the network attacks and defenses work
- Practical skills in demonstrating the network attacks and defenses
- Design logic, programming style, and readability of your program

Note: Your program should not be found from the Internet (i.e. NO plagiarism, do NOT simply copy and amend programs found from the Internet and submitted as yours.) In fact programs that are more complicated than necessary will get lower grades. So use this assignment as a learning opportunity to practice and write the tool yourself.